

XIN WANG

+1 (314) 668-0600 wxin@wustl.edu [website](#)

EDUCATION

Washington University in St. Louis

PhD student, Computer Science | GPA: 3.83

St. Louis, MO

Aug 2025 - May 2030 (expected)

New York University

MS, Computer Engineering | GPA: 3.933

New York, NY

Sep 2022 - May 2024

Xinjiang University

BE, Computer Software Engineering | GPA: 3.58

Urumqi, China

Sep 2017 - Jun 2021

RESEARCH EXPERIENCE

LLM-Driven User Profiling with Consistency Analysis

Washington University in St. Louis | Advisor: Prof. Umar Iqbal

Aug 2025 - Oct 2025

St. Louis, MO

- Investigated automated user attribute extraction from unstructured dialogue using GPT-5-mini, developing hierarchical classification AI agents for demographic and interest profiling across 200+ categories with taxonomy-guided prompts design.
- Established comprehensive evaluation methodology incorporating output stability metrics, cross-run similarity analysis (Jaccard coefficient), and soft consensus scoring.

Baseband Security and RIL Vulnerability Analysis

Dartmouth College | Advisor: Prof. Kyungtae Kim

Oct 2024 - Dec 2024

Remote

- Evaluated domains including Baseband, WebAssembly, Apple System Software, Windows System Software, and Rust, ultimately concentrating on Baseband security.
- Analyzed key papers—BASESPEC, BASECOMP, DoLTest, and BaseMirror—to understand program analysis methodologies and identify vulnerabilities within the Radio Interface Layer (RIL).

LoRa IoT Devices Fuzzing and Security Analysis

University of California, Riverside | Advisor: Prof. Zhaowei Tan

Feb 2024 - Jan 2025

Remote

- Researched LoRaWAN protocols and fuzzing techniques, with a focus on the **Firm-AFL** framework for ARM and MIPS-based IoT devices.
- Identified challenges with directly applying Firm-AFL to TrackerD's ESP32 architecture, including exotic architecture emulation and hardware-specific dependencies.
- Developed a custom fuzzing solution by integrating **AFL++** with **Ghidra**'s emulation framework to address architecture-specific issues, enabling efficient fuzzing for the Xtensa-based firmware.
- Modified memory, register configurations, and hook definitions to align with the firmware's memory layout, input handling, and calling conventions.

Secure Networking with TailScale and MITM Proxy

New York University | Coursework Lab

Sep 2023 - Dec 2023

New York, NY

- Set up and utilized **MITM** proxy to intercept and manipulate network traffic for security testing and analysis.
- Conducted practical exercises in network monitoring and intrusion detection, using **TailScale**'s exit node feature and transparent proxying capabilities.

Cryptographic Protocols and Heartbleed Vulnerability Analysis

New York University | Coursework Lab

Sep 2023 - Dec 2023

New York, NY

- Developed and tested cryptographic algorithms, including **RSA** key derivation, encryption, decryption, signing, and verification using Python.

- Exploited the Heartbleed vulnerability to understand data leakage risks and conducted hands-on remediation by updating the **OpenSSL** library.
- Analyzed and manually verified X.509 certificates, enhancing understanding of public key infrastructure and digital signatures.

WORK EXPERIENCE

Teaching Assistant

New York University | ECE-GY 6353 Internet Architectures & Protocols Lab

Sep 2023 - May 2024

New York, NY

- Served as a diligent grader for a core course at NYU, evaluating student assignments and providing constructive feedback to enhance their understanding and skills in the subject matter.
- Demonstrated strong communication and organizational skills as a grader, effectively collaborating with faculty to maintain grading consistency and ensure fairness in the evaluation process.

Software Development Engineer Intern

iFLYTEK Co., LTD

Sep 2020 - Apr 2021

Hefei, China

- Developed an **Automated Deployment Tool** incorporating **FreeMarker** and **SpringBoot**, enabling test engineers to download required **yaml** configuration files based on scenarios, reduced 30% manual setup time.
- Increased stability by 26% for the *Video Analysis Platform* service that processes the data of 80 million users, by utilizing **Redis**, **Kafka** and **Elasticsearch** to generate and transmit vehicle flow statistics to **microservices**.
- Saved 42% time spent on parameter verification and module package of *AI Knowledge Graph Platform* by introducing **POJO** extraction tool that simplifies the hierarchy of the objects.
- Reached 56% efficiency improvement of the *Shared Abilities Platform* that provides services of SDK or API callback, by dockerizing & optimizing the existing platforms that conditionally interact with.
- Created a web service using **Nginx**, created scripts to crawl the *Tongling City SuperBrain* online data with Python, cleansed and processed for analysis by **BeautifulSoup** with 60% less labor used.

Software Development Engineer Intern

Suncaper Data Co., LTD

May 2020 - Aug 2020

Chengdu, China

- Performed as a project manager and ensured the completeness of all works in each sprint by planning and following up with project status via **JIRA**.
- Conceptualized and implemented an E-Commerce Platform based on **SSM** architecture , ensured the fulfillment of comprehensive processes of shopping & item recommendations empowered with **Hadoop**, **Spark**.
- Constructed a **MySQL** database with **Power Designer**, using **Navicat** to create tables that prepared the data source for **MyBatis Generator** to generate **POJO** and **Example** instances for the initialization of the system.

AWARDS AND HONORS

- **Outstanding Graduates** | XJU
 - **Outstanding Graduation Design (Thesis)** | XJU
 - **Xinjiang Uygur Autonomous Region Encouragement Scholarship** | Top 3%
 - **Third Prize of Northwestern Division** | China Collegiate Computing Contest
- Jun 2021
- Jun 2021
- Nov 2020
- Aug 2020

SKILLS

Languages	Java, JavaScript, C++, Python, SQL, HTML5, C#, Bash/Shell scripting
Frameworks/Libraries	React.js, NodeJs, SpringBoot, Kafka, Hadoop, Spark
Development Tools	Git, Jira, Docker, Kubernetes, Postman, PowerDesigner
Database	MySQL, MongoDB, Oracle Database, Redis, MinIO